26900557 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

Regular EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEZGYA00 - CLINICAL GENOMICS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. The world's first baby born through IVF, Louise Brown, was born in the year [K] / [CO1]

   a). 1953 b). 1968 c). 1978 d). 1990

2. Liquid nitrogen used for cryopreservation maintains a temperature of about [K] / [CO1]

   a). 0 °C b). −20 °C c). −80 °C d). −196 °C

3. Characteristic "happy puppet" appearance with frequent laughter is seen in [K] / [CO2, CO3]

   a). Turner syndrome b). Down syndrome

   c). Angelman syndrome d). Prader-Willi syndrome

4. G6PD deficiency is detected mainly by [K] / [CO2, CO3]

a). urine culture b). ultrasound c). karyotyping d). an enzyme assay

5. TAC3 codes for [K] / [CO2, CO3]

a). kisspeptin b). GnRH receptor c). leptin d). neurokinin B

6. The four processes of pharmacokinetics are [K] / []

a). binding, folding, cutting, joining

b). synthesis, packaging, sale, use

c). absorption, distribution, metabolism, excretion

d). inhalation, digestion, storage, exhalation

7. Euchromatin is [K] / []

a). found only in prokaryotes

b). made of RNA only

c). tightly packed and inactive

d). loosely packed and transcriptionally active

8. A multifactorial disease is one caused by [U] / []

a). a chromosome number change only

b). infection alone

c). a single gene only

d). many genes together with environmental factors

9. A gene panel is useful when [U] / [CO4]

a). several genes can cause a similar phenotype

b). only one known mutation is suspected

c). the disease is purely infectious

d). no genetic test exists

10. Lynch syndrome is caused by mutations in [K] / [CO4]

a). the CFTR gene b). the HBB gene

c). DNA mismatch repair genes d). BRCA1 only

[1x10 = 10]

Part B

Short Answer Questions

Answer any **14** questions

Each question carries **2** marks

11. Write any two roles of a clinical geneticist. [K] / [CO4]

12. Define sensitivity and specificity of a test. [K] / [CO1]

13. What is gamete donation? [U] / [CO1]

14. What is oocyte retrieval? [U] / [CO1]

15. Why can PGT be done only along with IVF? [U] / [CO1]

16. What is whole-exome sequencing? [U] / [CO2, CO3]

17. Write any two roles of a genetic counselor in ART. [K] / [CO1]

18. What is chromosomal non-disjunction? [U] / [CO2, CO3]

19. Write any two clinical features of Turner syndrome. [K] / [CO2, CO3]

20. How does congenital bilateral absence of the vas deferens relate to the CFTR gene? [U] / [CO2, CO3]

21. Why may tamoxifen fail in a CYP2D6 poor metaboliser? [U] / []

22. What is a genetic polymorphism? [U] / []

23. Define epigenetics. [K] / []

24. Differentiate between a beta-thalassemia carrier and an affected individual in genetic terms. [U] / [CO4]

25. What is a pedigree chart? Write one use of it. [U] / [CO4]

26. Mention any two genetic analysis technologies used in PGT. [K] / [CO1]

[2x14 = 28]

Part C

Short Essay Questions

Answer any **8** questions

Each question carries **4** marks

27. Describe the counseling aspects to be covered before and after whole-exome sequencing in a child.	[U] / [CO5]
28. Describe the heel prick test and explain the principle and applications of tandem mass spectrometry in newborn screening.	[U] / [CO2, CO3]
29. Explain the technique of Gamete Intrafallopian Transfer with its indications.	[U] / [CO1]
30. Give an overview of the common neonatal genetic disorders with their clinical features.	[U] / [CO2, CO3]
31. Evaluate the statement: "Host genetics determines who develops active tuberculosis after infection." Support your answer with evidence.	[E] / []
32. Explain the principle and examples of epigenetic therapy in cancer treatment.	[U] / []
33. Explain how the results of a GWAS are interpreted, with reference to Manhattan plots and odds ratios.	[U] / []
34. Explain the management and surveillance options offered to a woman found to carry a BRCA1 mutation.	[U] / [CO4]
35. Compare chromosomal microarray and exome sequencing as approaches to identifying genetic causes of autism.	[An] / [CO4]
36. A couple has one child with cystic fibrosis and is planning another pregnancy. Work out the recurrence risk and outline the counselling and testing options you would offer.	[A] / [CO4]

[4x8 = 32]